Pure strategy Nash equilibrium cell shaded.

Party 1 \ Party 2	P	D
P	(α_1, β_2)	$(0, \alpha_2)$
D	$(\beta_1, 0)$	(η_1, η_2)

$P =$ cooperate, $D =$ Anticipate

$\alpha_i > \beta_i > \eta_i > 0,\ i \in \{1, 2\}$

FIGURE 4.4 State of Nature as Assurance Dilemma

Hobbes identifies two types of parties in the State of Nature. Parties of only one of these two types prefer to Anticipate against cooperators. I will follow Kavka's terminology and refer to parties of this aggressive type as *dominators*. The more "modest" parties are *moderates*.[41] Both types most fear exploitation and prefer peace over war. However, a moderate most prefers mutual cooperation, while a dominator most prefers to exploit cooperators. Kavka argues that when a moderate meets a dominator in the State of Nature their encounter has the structure of a game Kavka dubs the *Assurance Dilemma*.[42] Figure 4.4 summarizes an Assurance Dilemma where Party 1 is a moderate and Party 2 is a dominator.

The moderate has Stag Hunt preferences, while the dominator has Prisoner's Dilemma preferences, so the Assurance Dilemma is a hybrid of these two games. An Assurance Dilemma has a determinate outcome if both know their preferences and at least the moderate knows that they are engaged in an Assurance Dilemma and that they are both rational. For D is Party 2's strictly dominant choice, and Party 1, knowing this and knowing Party 2 is rational, concludes that her own choices are effectively between (P, D) and (D, D). Since Party 1 prefers (D, D) over (P, D), she chooses D. Hence (D, D) is the unique *rationalizable* strategy profile of the Assurance Dilemma; that is, (D, D) is the only strategy profile compatible with the parties having common knowledge of their rationality and the

41. Kavka (1986), p. 97. In his later 1989 essay, Kavka calls the dominators *glory seekers*.

42. Kavka (1989). In Taylor (1987), pp. 140–141, and its predecessor volume *Anarchy and Cooperation* (London: John Wiley and Sons, 1976), pp. 109–110, Michael Taylor considered briefly an Assurance Dilemma game as a model of Hobbesian State of Nature interaction without naming the game. However, Taylor preferred a Prisoner's Dilemma interpretation of the Hobbesian State of Nature because he believed that Hobbes' texts supported the view that Hobbes thought Anticipation is *every* party's strictly dominant strategy in the State of Nature (1976, pp. 110–111, 1987, pp. 129–141). Kavka developed his Assurance Dilemma analysis of Hobbesian State of Nature interaction independently of Taylor's earlier work.

payoff structure of their game.[43] Kavka calls this game the Assurance *Dilemma* because (D,D) is the unique Nash equilibrium of this game even though only one of the parties prefers to deviate from the optimal (P,P) outcome. Kavka maintains that the Assurance Dilemma is

> the best simple representation of Hobbes's analysis of the problem of anarchy. Since it relies on less cynical assumptions about human nature than the standard prisoner's dilemma view, we may call this the *Kinder-Gentler* Interpretation of Hobbes's state of nature.[44]

The Assurance Dilemma model incorporates preferences that are "kinder and gentler" than Prisoner's Dilemma preferences. Still, if encounters in the Hobbesian State of Nature have an Assurance Dilemma preference structure, one can predict that rational parties with sufficient mutual knowledge of their situation will go to war. Hobbes' own texts indicate that he attributes "kinder, gentler" preferences to people than do many of his followers. If only all in the State of Nature were moderates, then peace would be an equilibrium. Alas, this is not likely to be the case. Dominators are likely to be present in the State of Nature, and these dominators will start a spiral of expected Anticipation that leads all to go to war.[45] The Assurance Dilemma illustrates a simple case of how this spiral unfolds.

Kavka's Assurance Dilemma model is an important step forward in the analysis of the Hobbesian State of Nature, but it has its shortcomings. The Assurance Dilemma model summarizes a compelling argument for why mutual Anticipation is inevitable when a rational moderate and a rational dominator meet and the moderate recognizes the dominator as such. But how often does this happen? When dominators meet, their encounter is a Prisoner's Dilemma.

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43. Doug Bernheim and David Pearce introduced the rationalizability concept in B. D. Bernheim, "Rationalizable Strategic Behavior," *Econometrica* 52 (1984), pp. 1007–1028, and David Pearce, "Rationalizable Strategic Behavior and the Problem of Perfection," *Econometrica* 52 (1984), pp. 1029–1050. If a game has a unique rationalizable strategy profile, as do the Prisoner's Dilemma and the Assurance Dilemma, then this profile is the unique Nash equilibrium of this game.

Kavka (1989, pp. 9–10) gives a similar argument that (A,A) is the determinate solution of the Assurance Dilemma. In his argument, Kavka does not state the necessary assumptions regarding the parties' knowledge of their situation. Also, Kavka was evidently unaware that his argument is an informal rationalizability argument.

44. Kavka (1989), p. 10. I have followed the exact punctuation, capitalization, and emphases in Kavka's unpublished manuscript.

45. Kavka (1986), §3.4–6, (1989).

When moderates meet, their encounter is a Stag Hunt. If moderates can identify the dominators in their midst, then of course they should be able to identify the other moderates. Such moderates might adopt the contingency strategy of cooperating with moderates and Anticipating against dominators. The premises necessary to enable one to predict that parties in an Assurance Dilemma will surely follow (D, D) on account of their rationality also imply that when moderates are present in a Hobbesian State of Nature, the outcome need not be Hobbes' war of all against all. Indeed, if the moderates in anarchy consistently cooperate with the other moderates and Anticipate against the dominators, the outcome is the intermediate state between peace and war that Simmons argues Locke predicted![46]

Kavka may not have fully recognized this problem for his Assurance Dilemma model. But he was evidently aware that not all encounters in anarchy will be Assurance Dilemmas. Kavka argues that the moderate parties in a Hobbesian State of Nature will reason *as if* they were in an Assurance Dilemma because these parties in fact *cannot* distinguish the dominators from the moderates. Given that the moderates are uncertain as to whether or not the other parties they meet are dominators, Kavka argues that their best choice is to play it safe and Anticipate. Consequently, these cautious moderates together with the dominators generate more expected Anticipation in an ever-increasing spiral.[47] Again, Hobbes himself gives a similar argument:

> we cannot tell the good and the bad apart, hence even if there were fewer evil men then good men, good, decent people would still be saddled with the constant need to watch, distrust, anticipate and get the better of others, and to protect themselves by all possible means.[48]

Hobbes, and Kavka after him, maintains that the moderates' uncertainty as to whether or not other parties they encounter are dominators ultimately leads the moderates to avoid ever running the risk of being exploited. Following their reasoning, a moderate will treat any other party she encounters as if she were certain that this other party is a dominator. Hence, from the perspective of a moderate, anarchy does have an Assurance Dilemma structure in which one's only rational alternative is to Anticipate.

Hobbes, and later Kavka, concludes that the parties' uncertainty regarding party types they meet is a key reason the Hobbesian State of Nature converges to

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46. Simmons (1989), (1993), ch. 1.

47. Kavka (1986), pp. 104–107, (1989).

48. *De Cive* Preface 12.